

Individual Addendum Template

QM 2023 Capstone Project - Milestone 4

Individual Contribution Statement Name: ___Katie Koonts___ Team: _Stats
2___ Date: ___April 30,2026___

1. Personal Contribution to Capstone Milestones

Milestone 1: Data Pipeline (Week 5)

Tasks completed:

Helped document the data cleaning rules for the CoinGecko and macro series pipeline, including merge logic, missing value handling, and variable definitions used in the data quality report and data dictionary

Hours spent: ___1.5___ hours

Milestone 2: EDA Dashboard (Week 10)

Tasks completed:

I did not contribute to milestone 2 because I was gone due to track, so I was unable to do anything for this portion of the project.

Hours spent: ___0___ hours

Milestone 3: Econometric Models (Week 14)

Tasks completed:

Worked on Part A of M3 by verifying the fixed effects Model A outputs and diagnostics used in the memo, including the heteroskedasticity discussion, VIF caveat, and robustness references

Hours spent: ___2___ hours

Milestone 4: Final Investment Memo (Week 14)

Tasks completed:

Drafted the methodology section for the final submission, prepared submission support text, and checked that the narrative matched the exported tables and figures.

Hours spent: __1.5__ hours

Total Estimated Contribution

Total hours across all milestones: __5__ hours

Percentage of team workload: __15__% (must sum to 100% across all team members)

Role(s) on team: Data documentation, results verification, methodology writing, and submission support

2. One Defended Methodological Decision

Decision: I supported using a two way fixed effects specification with clustered standard errors at the token level, rather than relying on a simple pooled regression

Reasoning: The panel has repeated observations for the same tokens over time, so token level clustering is necessary once the Breusch Pagan test and other diagnostics point to heteroskedasticity. Two way fixed effects also fit the design of our SEC event question because date fixed effects absorb market wide shocks, leaving the interaction terms to identify the relative exposure of token groups. A pooled model would have been easier to estimate, but it would have been less credible for this data structure.

Data evidence: In the final results, the clustered FE model shows a clear stablecoin event effect (coefficient = 0.007431, $p < 0.001$) while the DeFi effect is small and not statistically significant at baseline (coefficient = 0.002691, $p = 0.453$), which supports using a specification that isolates group level exposure.

Economic reasoning: The panel has repeated observations for the same tokens over time, so token level clustering is necessary once the Breusch Pagan test and other diagnostics point to heteroskedasticity. Two way fixed effects also fit the SEC event design because date fixed effects absorb market wide shocks, leaving the interaction terms to identify relative exposure by token group, which is the economically relevant question.

Robustness: The main interpretation is consistent with the robustness checks we reported, including lag structures, outlier trimming, and subgroup comparisons, so the result is not driven by one narrow specification.

Alternative considered (and why rejected): I considered a pooled regression without date fixed effects, but rejected it because it would confound SEC event effects with broader market moves and would weaken the credibility of the comparison across token groups.

3. One Key Limitation of Our Analysis

Limitation: The model has severe multicollinearity between the size and liquidity controls, especially market cap and trading volume.

Why this matters: That makes the individual control coefficients unstable, so the exact magnitude on each control should not be overinterpreted even when the overall fit looks reasonable.

Potential mitigation: In future work, we could reduce overlap by orthogonalizing the controls, using principal components, or testing a more compact specification that focuses on the core event and group effects.

Example: For instance, when both market capitalization and trading volume are included the market cap coefficient flips sign and its standard error increases across robustness checks, illustrating how correlated controls can make individual estimates unstable.

5. Self-Reflection

What did I do particularly well on this capstone?

I stayed focused on making the project documentation consistent with the actual analysis outputs.

Example: I systematically cross checked exported regression tables against plotted figures and corrected captions and labels so the memo accurately reflected the results.

What could I have improved?

I could have started the final memo synthesis earlier so there would have been more time for revision.

Example: I could have begun drafting the memo two weeks earlier and scheduled weekly checkpoints with teammates to catch inconsistencies sooner.

What did I learn from this capstone project?

I learned how important it is to tie written interpretation back to the underlying data workflow and model output instead of treating the memo as a separate task. I also learned that clear documentation is part of the analysis, not just an afterthought.

Example: learned that recording data cleaning decisions early and running quick diagnostics (e.g., VIF, Breusch Pagan) prevents last minute model revisions and improves interpretability.

6. Attestation

By submitting this individual addendum, I affirm that:

All contributions listed above are accurate and honest

I have not exaggerated my role or minimized teammates' contributions

I understand that this addendum may be used to adjust my individual grade relative to the team
grade

I take full responsibility for my work and any errors in the sections I authored

Signature: _____Katie Koonts_____ **Date:** ____04/30/2026_____

Submission Instructions